A PROJECT SYNOPSIS

ON

AI-Based Underwater Monitoring, Image Enhancement, Detection, and Deep Mapping

FOR THE DEGREE OF

BACHELOR OF ENGINEERING IN

COMPUTER ENGINEERING

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Abstract

This project presents an AI-based system for underwater monitoring, image enhancement, garbage detection, and deep mapping. It addresses the growing issue of underwater pollution in places such as rivers, ponds, and lakes—particularly in zones where human access is unsafe or ineffective due to poor visibility or water depth. The system automatically enhances underwater images to improve clarity, detects and classifies waste materials using deep learning models like YOLOv8, estimates the garbage density in each area, and maps the depth of polluted zones using advanced computer vision techniques

The solution targets the increasing issue of underwater pollution in water bodies like rivers, lakes, and ponds—especially in zones that are difficult or dangerous for humans to access. The system captures real-time underwater images or videos, enhances them for clarity, detects and classifies waste such as plastic, cans, or organic material using deep learning (YOLOv8), and estimates the depth and density of each polluted zone. Based on this data, the system classifies zones as clean, moderate, or critical and visualizes everything on a dashboard for actionable insights.

This system helps environmental authorities plan effective cleaning operations using data rather than guesswork, making underwater cleanup safer, faster, and smarter.The system works by first enhancing underwater footage using advanced computer vision algorithms to make murky or distorted visuals clearer. Then, it detects and classifies garbage using deep learning (YOLOv8) and calculates the depth and density of the trash to categorize zones based on severity. This makes it possible to determine the best cleanup strategy—manual, robotic, or deferred.

Keywords

Underwater monitoring, AI-based detection, image enhancement, YOLOv8, GAN, MiDaS, trash density, zone classification, environmental AI, Streamlit dashboard.

Introduction

Underwater environments have become increasingly polluted with garbage, plastics, and debris, which severely impact marine life and water quality. Cleaning such underwater zones is a challenge due to poor visibility, uncertain depth, and health hazards for human divers. To solve this, we propose an AI-powered system that automates the detection and classification of underwater trash, enhances visibility in murky waters, and maps garbage-heavy zones using depth estimation models. Using a combination of computer vision and deep learning, the system processes underwater images in real time and displays information about trash types, depth, and risk level through a centralized dashboard

Water pollution has become an alarming issue globally, and in many cases, garbage accumulates in water bodies in areas that are hard or unsafe to reach. Often, cleanup efforts are limited to visible surface waste, leaving underwater trash untouched due to poor visibility or depth. This creates long-term ecological problems, such as harming marine life or polluting drinking water sources.

Manual cleaning of underwater environments is risky, slow, and expensive. Divers cannot always assess what lies beneath or how safe it is to clean certain zones. Moreover, without real-time data or visuals, cleanup authorities often guess where to start, which results in inefficient resource use and time loss. This project is designed to solve those challenges using automation and AI.

By leveraging image processing, deep learning, and dashboard visualization, this project introduces an AI system that processes underwater images in real time. It detects, classifies, and maps trash-heavy areas based on depth and density. The output is presented visually on a dashboard, helping users make informed decisions quickly and safely.

Objective

- To develop a system that captures underwater images or videos using real-time or offline sources (e.g., cameras, recorded footage).
- To enhance underwater images using deep learning models like U-Net, Water-Net, or DehazeNet for better clarity and contrast.
- To improve image visibility in murky or low-light conditions using advanced image preprocessing techniques.
- To detect and classify underwater garbage items (e.g., bottles, bags, cans, organic waste) using YOLOv8 or YOLOv7t-CEBC object detection models.
- To train or fine-tune garbage detection models using custom or publicly available underwater datasets.
- To estimate the depth of each object in the image using MiDaS v3.1 (monocular depth estimation) without requiring special hardware.
- To calculate trash density in each frame based on the percentage of area covered by detected waste.
- To create rules or thresholds to label zones as Clean, Moderate, or Critical based on trash density and depth information.
- To visualize all the processed data (original frame, enhanced frame, detected objects, depth map, etc.) on a single interactive dashboard.
- To build an intuitive and lightweight Streamlit dashboard that can be used by non-technical users.
- To include interactive charts like bar, pie, or line graphs to show distribution of garbage types and density levels.
- To display zone-wise suggestions for cleanup actions (e.g., manual cleaning, robotic vacuum, or skip area due to danger).
- To allow users to upload video or images, process them, and download processed outputs like CSV reports and annotated images.
- To automate the entire pipeline with minimal human intervention — from input capture to visual output.

Motivation

In many water bodies, especially in developing regions, manual underwater garbage removal is dangerous, inefficient, and often ineffective due to limited visibility, unknown water depth, and the uncertain type of waste present. Human divers face risks from contaminated water, sharp materials, or entanglement in underwater debris. The motivation behind this project is to build a smart, automated solution that reduces human risk while increasing the effectiveness of trash detection and removal. With AI, this system can identify polluted zones, classify risk levels, and suggest the right approach—robotic, manual, or combined—thus supporting cleaner, safer, and more sustainable water bodies.

One of the biggest motivations for this project is the rising pollution in urban and semi-urban water bodies. These places are often filled with waste, and it's dangerous or impractical to send divers without knowing what lies underwater. The idea is to build a system that eliminates guesswork and risk.

Many cleanup projects are delayed because authorities lack real-time data or cannot visually inspect submerged zones. With better monitoring tools powered by AI, teams can act faster and with more confidence. This project gives them the vision and analysis needed to operate more efficiently.

Literature Survey

Recent research shows a growing interest in AI for underwater trash detection. For instance, the YOLOv7t-CEBC paper introduces a compressed version of YOLO optimized for real-time underwater detection, achieving both speed and accuracy. It's ideal for deployment in resource-limited systems, just like in your project.

The IBURD project emphasizes the use of synthetic underwater datasets to overcome the lack of labeled data. This is important because underwater datasets are rare and expensive. By mixing real backgrounds with object masks, researchers created training material that improves AI performance in murky conditions.

Another model, BG-YOLO, combines enhancement and detection pipelines for better clarity and accuracy in poor visibility. Likewise, the LUIEO paper proposes using physics-inspired models for underwater light distortion to achieve better detection even in harsh conditions. These studies validate the effectiveness of your project design and provide references for improving its accuracy and robustness.

Recent research shows the growing interest in AI-based underwater detection. A paper on YOLOv7t-CEBC (2024) presents an enhanced version of YOLOv7-tiny optimized for underwater trash detection with real-time capability and high accuracy. Another study**,** IBURD (2025)**,** proposes generating synthetic underwater datasets by blending backgrounds and objects to improve model training in murky environments. BG-YOLO (2024) combines image enhancement and garbage detection into a single pipeline for better performance in low-visibility scenes. AASS + SRR + YOLOv8 (2024) uses super-resolution and aerial scanning to detect surface and shallow underwater trash more accurately. Lastly, LUIEO (2024) offers a unified AI model for joint underwater enhancement and detection, using light propagation and haze modeling to reconstruct clearer underwater visuals. These research efforts highlight the potential and evolving efficiency of combining detection with enhancement in underwater applications.

Problem Statement

To design and implement a real-time AI system that enhances underwater images, detects and classifies trash, estimates the depth and density of pollution, and categorizes underwater zones based on risk level. The system must be capable of working in real-time scenarios using camera feeds or videos and should present this information on a user-friendly dashboard. The goal is to reduce manual effort, improve safety, and support environmentally responsible underwater waste management.

Conclusion

This project leverages the power of artificial intelligence to address real-world environmental challenges in underwater garbage detection and management. By combining image enhancement, deep learning-based detection, and depth estimation, the system offers a complete pipeline for monitoring and analyzing underwater zones. Its ability to classify trash-heavy areas into zones and display insights through a dashboard makes it practical for municipal corporations, NGOs, and environmental conservation teams. The project not only improves the efficiency and safety of underwater cleanup but also contributes to the broader goal of sustainable and smart environmental practices.

This project successfully combines AI technologies to automate underwater monitoring, garbage detection, and deep mapping. From capturing raw footage to generating real-time analytics and actionable insights, the system covers the full lifecycle of underwater waste assessment.

By using YOLOv8 for detection, MiDaS for depth estimation, and Streamlit for dashboard visualization, the project achieves a high-tech but low-cost solution that can be deployed in real-world scenarios. It is especially valuable for areas with limited access to expensive drones or sonar equipment.

Overall, this system promotes sustainable water management, supports environmental cleanup efforts, and opens doors to smart, AI-driven public services. It aligns with goals like “Clean Water and Sanitation” and “Sustainable Cities and Communities,” making it not just technically sound but socially impactful.

References

1. Chaudhary, R., Choudhury, A., & Sinha, A. (2018). *Underwater Image Enhancement Using Deep Learning*. In IEEE International Conference on Computer Vision and Image Processing.
2. Islam, M. J., Xia, Y., & Sattar, J. (2020). *Fast Underwater Image Enhancement for Improved Visual Perception*. IEEE Robotics and Automation Letters, 5(2), 3227–3234.
3. Zhao, X., Li, Y., & Gong, D. (2020). *A Real-Time Underwater Object Detection Algorithm Using YOLOv3*. Sensors, 20(6), 1613.
4. Song, W., Bai, X., & Zhang, Y. (2021). *Underwater Object Detection Using Improved YOLO Algorithm Based on Attention Mechanism*. Journal of Marine Science and Engineering, 9(5), 506.
5. Li, C., Guo, J., & Cong, R. (2021). *Underwater Image Enhancement via Medium Transmission-Guided Multi-Color Space Embedding*. IEEE Transactions on Image Processing, 30, 4985–4999.
6. Islam, M. J., Xie, Z., & Sattar, J. (2022). *Semantic Segmentation-Based Detection of Marine Debris in Underwater Videos*. Robotics and Automation Letters, 7(2), 2515–2522.
7. MiDaS Team (2022). *MiDaS v3: Monocular Depth Estimation Using Vision Transformers*.
8. Gajjar, R., & Shah, S. (2023). *YOLOv7-tiny with Channel and Spatial Attention for Underwater Garbage Detection (YOLOv7t-CEBC)*. Preprint on arXiv:2305.10934.
9. Chen, Y., Li, X., & Zhang, H. (2023). *IBURD: Improved Blending-Based Underwater Real-Time Dataset Generation for Object Detection*. Proceedings of ICPR 2023.
10. Ahmed, F., & Zhou, D. (2024). *LUIEO: Light-Propagation-Aware Underwater Image Enhancement and Object Detection in End-to-End Networks*. IEEE Transactions on Neural Networks and Learning Systems.